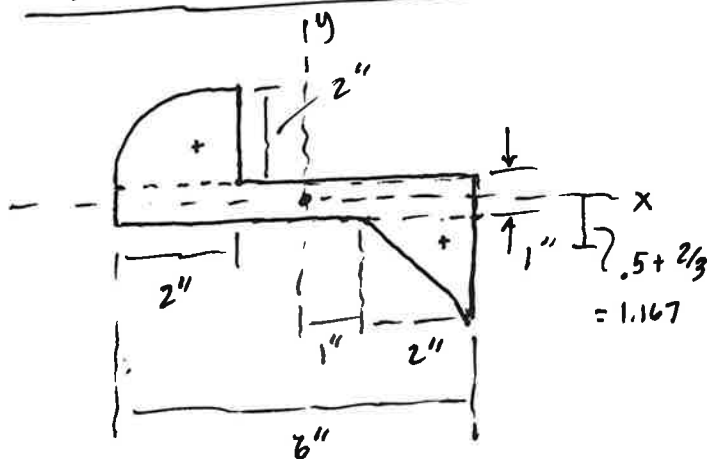


Ex Problem



$$\frac{4R}{3\pi} = \frac{4(2)}{3\pi} = 0.8488 \text{ in}$$



$$A = \frac{\pi(2)^2}{4} = \pi \text{ in}^2$$

$$\bar{I} = .05488 R^4$$



$$A = \frac{1}{2}(2)(2) = 2 \text{ in}^2$$

$$\bar{I} = \frac{1}{36}(b)(h^3) = \frac{2^4}{36}$$

$$I_x = \frac{1}{12}(6)(1)^3 + \left[.05488(2)^4 + \pi \left(.5 + \frac{2}{3} \right)^2 \right] + \left[.4444 + 2 \left(.5 + \frac{2}{3} \right)^2 \right]$$

$$= .5 + 6.593 + 3.167$$

$$I_x = 10.26 \text{ in}^4$$

$$I_y = \sum (\bar{I} + Ad^2) = \left[\frac{1}{12}(1)(6)^3 \right] + \left[.05488(2)^4 + (\pi)(1 + .8488)^2 \right] + \left[.4444 + 2 \left(3 - \frac{2}{3} \right)^2 \right]$$

$$= 18 + 11.616 + 11.33$$

$$I_y = 40.95 \text{ in}^4$$

$$I_{xy} = \sum [\bar{I}_{xy} + Adx dy]$$

$$= [0 + 0] + \left[+.01647(2)^4 + \pi(-1.8488) \left(.5 + \frac{2}{3} \right) \right] + \left[-\frac{1}{12}(2)^2(2)^2 + 2(+2.33)(-1.167) \right]$$

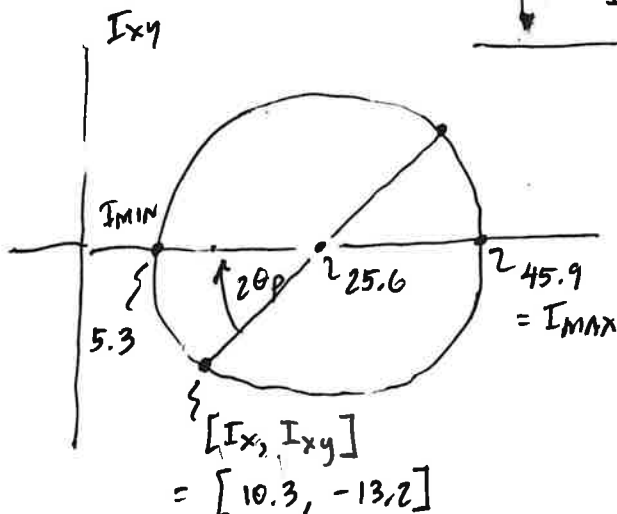
$$I_{xy} = -7.57 - 5.66$$

$$I_{xy} = -13.23 \text{ in}^4$$

$$I_{AVG} = \frac{I_x + I_y}{2}$$

$$= 25.6 \text{ in}^4$$

$$Radius = 20.3 \text{ in}^4$$



$$I_{MAX} = 25.6 + 20.3 = 45.9$$

$$I_{MIN} = 25.6 - 20.3 = 5.3$$

$$2\theta_p = -40.8^\circ$$

$$\theta_p = -20.4^\circ$$

